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Network Traffic Monitor

Project Report

1. Introduction

This project is a web-based simulation of a network traffic monitoring system developed using HTML, CSS, and JavaScript. It generates and displays packets in real time for analysis.

2. System Architecture

The project follows a modular structure:

• index.html – User interface

• data.js – Dataset file

• app.js – Logic and functions

3. Application State

The system uses a centralized object 'appState' to store packets, counters, interval control, and running status.

4. Core Functions

• buildPacket() – Generates simulated packets

Creates a packet object by selecting random protocol, IP addresses, ports, and size from the dataset defined in `data.js`.
Returns the generated packet which is later stored and displayed.

• appendRow() – Displays packets in table

Dynamically creates a table row and inserts packet data into the Packet Log table.
Removes the initial "No packets captured yet" row when first packet arrives.
Stores protocol, source IP, and destination IP in dataset attributes for filtering.

• logActivity() – Logs system actions

Adds a new log entry in the Activity Log section with the current time.
Displays messages such as start, stop, filter, and clear actions.
New logs are inserted at the top for better visibility.

• updateStats() – Updates statistics

Counts total packets and categorizes them into TCP, UDP, and ICMP. Calculates average packet size using total size divided by number of packets. Updates all statistic values in the dashboard UI.

• startMonitoring() – Starts packet generation

• stopMonitoring() – Stops monitoring

• clearAll() – Clears all data

• applyFilter() – Applies filtering

• resetFilter() – Resets filters

5. Filtering Mechanism

Filters are applied using protocol, source IP, and destination IP. Rows are shown or hidden dynamically based on matching conditions.

6. Statistics

The system calculates total packets, protocol distribution, and average packet size.

Average Size = Total Packet Size / Total Number of Packets

7. Features

• Real-time packet simulation

• Dynamic UI updates

• Filtering system

• Activity logging

• Statistics computation

- # Scope Of Project

This project focuses on the design and implementation of a **network traffic monitoring simulation system**. It covers packet generation, visualization, filtering, and statistical analysis within a browser-based interface.

The scope includes:

- Simulation of network packets using predefined datasets (protocols, IPs, ports, services)
- Implementation of filtering mechanisms based on protocol, source IP, and destination IP
- Calculation of basic statistics such as total packets, protocol distribution, and average packet size

- Activity logging for monitoring user interactions (start, stop, filter, clear)

The system is limited to **simulated packet data** and does not perform actual network packet capture. Advanced features such as deep packet inspection, intrusion detection, and real-time hardware-level monitoring are outside the scope of this project

8. Conclusion

This project demonstrates the working of a basic network monitoring system and provides a strong understanding of packet flow and analysis.